

## **Serverless Architectures TASK – 1**

**1. Sign up for a free AWS account (if you don't have one), log in to the AWS Management Console, and navigate to the AWS Lambda service dashboard.**

**Ans :**

- Open AWS Free Tier and select Create a Free Account if you don't already have an AWS account.
- Complete the AWS sign-up process, including email verification, account information, payment method, and identity verification.
- Once your account is activated, open the AWS Management Console and sign in.
- In the AWS Console search bar, type Lambda.
- Select Lambda under Services.
- You should now be on the AWS Lambda dashboard, where you can create and manage serverless functions.

**2. Create a new AWS Lambda function named InstaLikeCounter that takes an event with a 'likes' number and returns the value increased by 1, simulating an Instagram-like button click.**

**Ans :**

1. Open <https://console.aws.amazon.com/> and sign in to your AWS account.

2. In the AWS Console search bar, type "Lambda" and open the Lambda service.

3. Click "Create function".

4. Select "Author from scratch".

5. Enter the function name:

InstaLikeCounter

6. Select "Python 3.x" as the Runtime.

7. Under Permissions, keep the default option to create a new execution role with basic Lambda permissions.

8. Click "Create function".

9. In the Code source editor, replace the existing code with:

```
def lambda_handler(event, context):  
    likes = event.get("likes", 0)  
    return {  
        "likes": likes + 1  
    }
```

10. Click "Deploy".

11. Click "Test" and create a new test event with this JSON:

```
{  
  "likes": 10  
}
```

12. Save the test event and click "Test" again.

13. Verify that the response is:

```
{  
  "likes": 11  
}
```

14. Test with different values, for example:

```
{  
  "likes": 99  
}
```

15. The Lambda function should return:

```
{  
  "likes": 100  
}
```

}

**3. Test your InstaLikeCounter Lambda function using the built-in AWS Lambda test feature by providing a sample event with 'likes': 99, and verify the output shows 100.**

**Ans :**

- Open the AWS Management Console:
- <https://console.aws.amazon.com/>
- Search for "Lambda" and open the AWS Lambda service.
- Click the function named "InstaLikeCounter".
- Open the "Test" section.
- Create a new test event.
- Enter this JSON:
  - {
  - "likes": 99
  - }
- Give the test event a name, for example:
- LikeTest
- Click "Save".

- Click "Test" to run the Lambda function.
- Check the execution result.
- Verify that the response is:
  - {
  - "likes": 100
  - }
- If the output shows "likes": 100, the InstaLikeCounter Lambda function is working correctly.

**4. Write a short comparison (3-4 lines each) of AWS Lambda, Azure Functions, and Google Cloud Functions, focusing on how each lets you run code without managing servers.**

**Ans :**

### **AWS Lambda**

Runs your code in response to events without requiring you to provision or manage servers.

AWS automatically handles infrastructure, scaling, and execution capacity.

You pay based on the compute resources and execution time your functions use.

### **Azure Functions**

Microsoft's serverless platform for running event-driven code without managing servers.

It integrates closely with Azure services such as Event Grid, Storage,

and Cosmos DB.

Azure automatically scales function instances based on incoming workload.

## **Google Cloud Functions**

Google's serverless service for executing code in response to events or HTTP requests.

You deploy individual functions while Google manages the underlying infrastructure and scaling.

It integrates naturally with Google Cloud services such as Cloud Storage and Pub/Sub.

**5. Use the AWS Lambda code editor to add a console.log statement that prints 'Function executed!' every time your function runs.<br><br><em><strong>Hint:</strong> Check the Monitor tab after running your test to see the log output.</em>**

**Ans :**

- Open the AWS Management Console:
- <https://console.aws.amazon.com/>
- Search for "Lambda" and open the Lambda service.
- Select your function:
- InstaLikeCounter
- In the Code tab, find the existing Lambda code.
- Update it to:

- `exports.handler = async (event) => {`
- `console.log("Function executed!");`
- `const likes = event.likes || 0;`
- `return {`
- `1. likes: likes + 1`
- `};`
- `};`
- Click "Deploy" to save the updated function.
- Click "Test".
- Use this test event:
- `{`
- `"likes": 99`
- `}`
- Run the test. The function should return:
- `{`
- `"likes": 100`
- `}`

- Open the "Monitor" tab.
- Click "View CloudWatch logs" or open the latest log stream.
- Find the log entry:
- Function executed!
- This confirms that the `console.log` statement runs every time the Lambda function is invoked.